PROJECT 1

SECTION A

QUESTION 1

A. Some important factors to consider are:

The legal and compliance requirements, this is to ensure that legal and regulatory compliance, the standards should assist the business in adhering to regional, governmental and global rules.

The coverage and scope, this entails that the wide range of topics related to threat management, incident response, data protection and access control should be covered by the security standards.

Scalability and flexibility, the chosen standards ought to evolve with the company enabling it to adjust as it grows or encounters new threats.

The industry relevance, these standards ought to be in line with the particular security and legal requirements of the sector in which the business works.

B. Legacy systems may experience performance overhead when implementing current security measures due to outdated hardware and software that cannot handle new encryption or logging needs efficiency

There might be some complicated integration, whereby integrating new security standards with outdated systems may necessitate additional middleware and potentially increase costs and maintenance overhead.

Legacy systems may not support modern security standards including protocols encryption and authentication techniques.

Older systems may not receive security updates resulting in vulnerabilities that modern security standards cannot address without disrupting operations.

C. This might bring additional risk to the organization by:

Cultural integration challenges, during integration stage operational misalignments that result in gaps in security controls can be caused by differences in the security cultures and risk management strategies of the two firms.

People from the same business may be considered insider threats either on purpose or accidentally.

The smaller business might handle data differently or might not follow the same rules as Delware Inc.

The network of Delware Inc, may become vulnerable due to the acquired company's possible laxer security regulations or insufficient protections.

D. Configuration, setting up the SIEM to correctly corelate data from various systems can be challenging and need specific knowledge to adjust the system to prevent false negatives or positives which could lower the efficiency of the security plan.

Combining data from several sources might result in excessive volumes of data which can cause SIEM's ability to efficiently interpret and rank alerts sometimes e\leading to missing reactions to serious threats.

The integration and correlation of data in the SIEM may become more difficult if various vendors employ distinct data formats and logging structures. Because of protocols the IDS and IPS systems might not be able to communicate with one another without the need for extra tools.

SCENARIO 2

1.2 A, Advantages of using IPv6 over IPv4 are,

IPv6 has greater address space compared to IPv4. IPv6 has 128 bits addressing allowing multiple of devices to connect without experiencing address depletion.

Compared to IPv4 which offers IPsec as an option optional feature, IPv6 was constructed with IPsec as a standard feature.

IPv6 increases packet processing efficiency and decreases size of routing tables.

Networking configuration is made simpler with IPv6 support for configuration which does away with the requirement for DHCP.

B. Man in the Middle attacks can intercept communication between the authentication server and the client if mutual authentication is not implemented. Authentication protocol weakness is one of the security issues you need to consider.

Misconfigured switches that improperly enforce port-based authentication are one example of how improperly configuring 802.1x could lead to security flaws.

C. If appropriate routing protocols are not used there may be an increase in latency or packet overhead due to increasing number of pathways between nodes.

There may also be issues with cost and deployment time if switching to a mesh network necessities new hardware and infrastructure.

Mesh arrangements networking allows for man pathways between devices and a single point of failure will not bring down the entire network resulting in high fault tolerance.

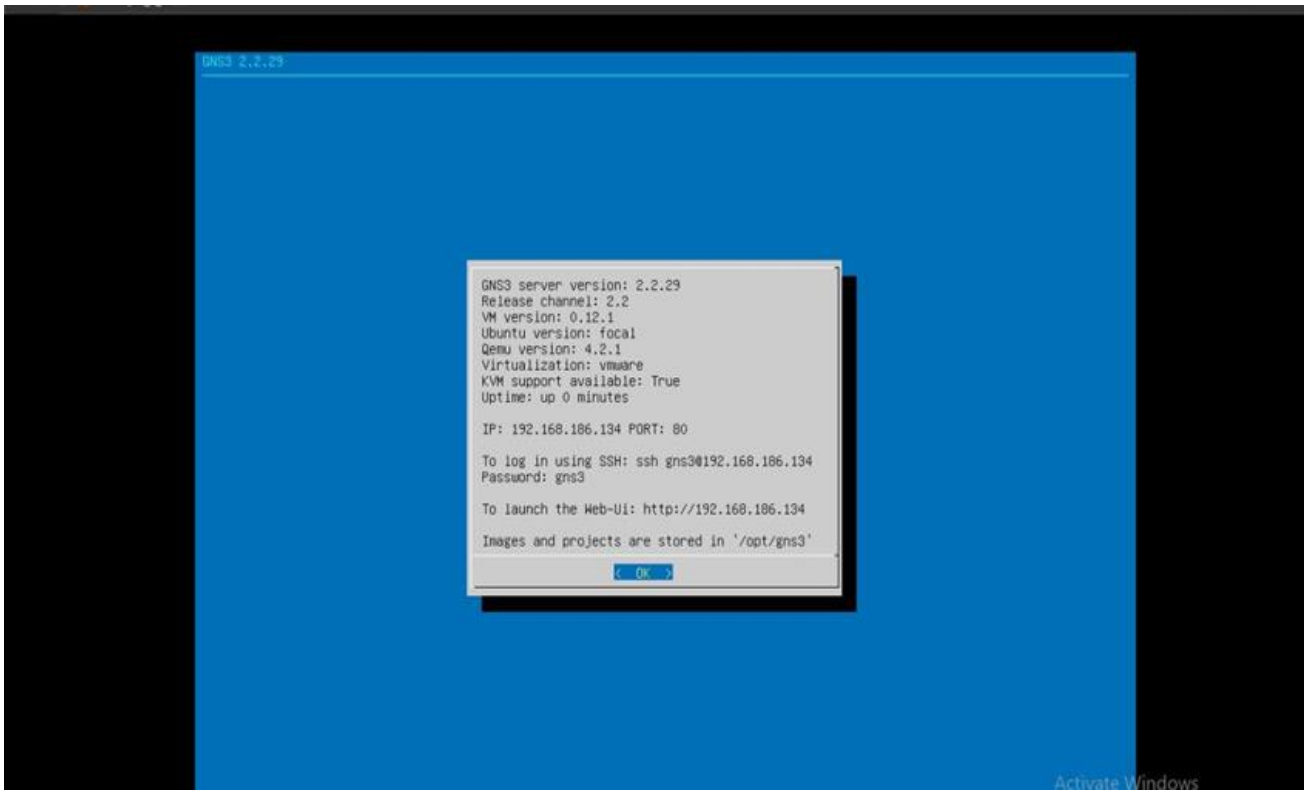
D. An essential component of the network architecture is the SDN an attacker might control the entire network if they manage to get access to the controller.

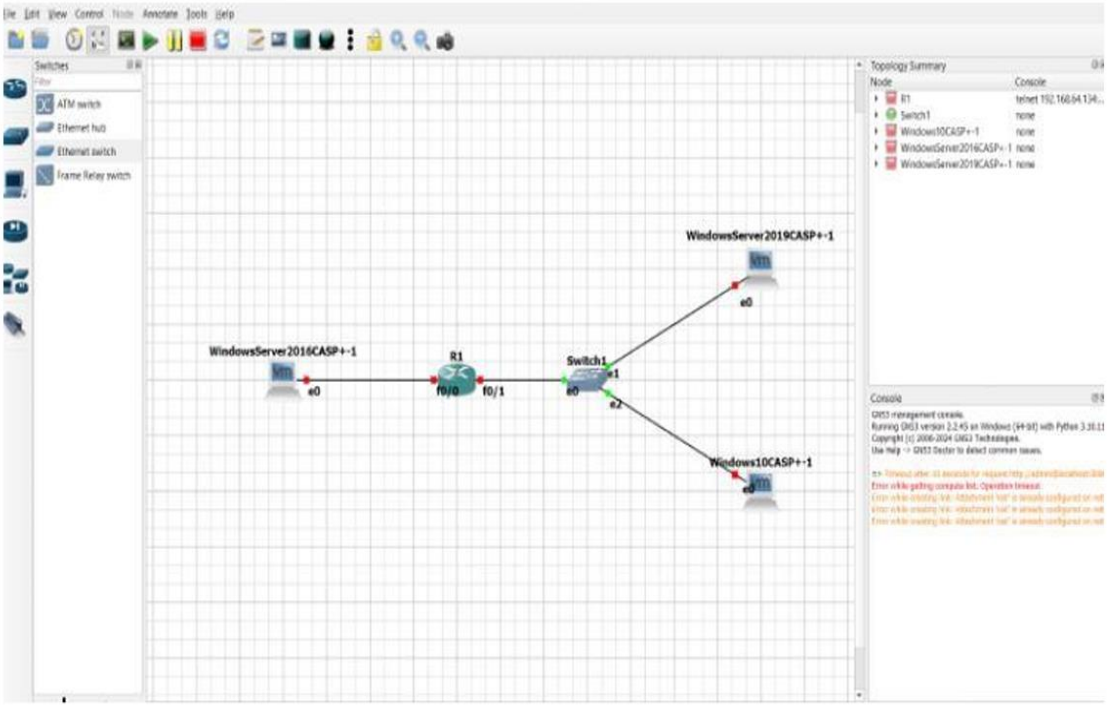
Data privacy concerns, the SDN is centralized, if appropriate encryption and access restrictions are not in place when interacting between the controller and the network devoices then sensitive network data may be exposed.

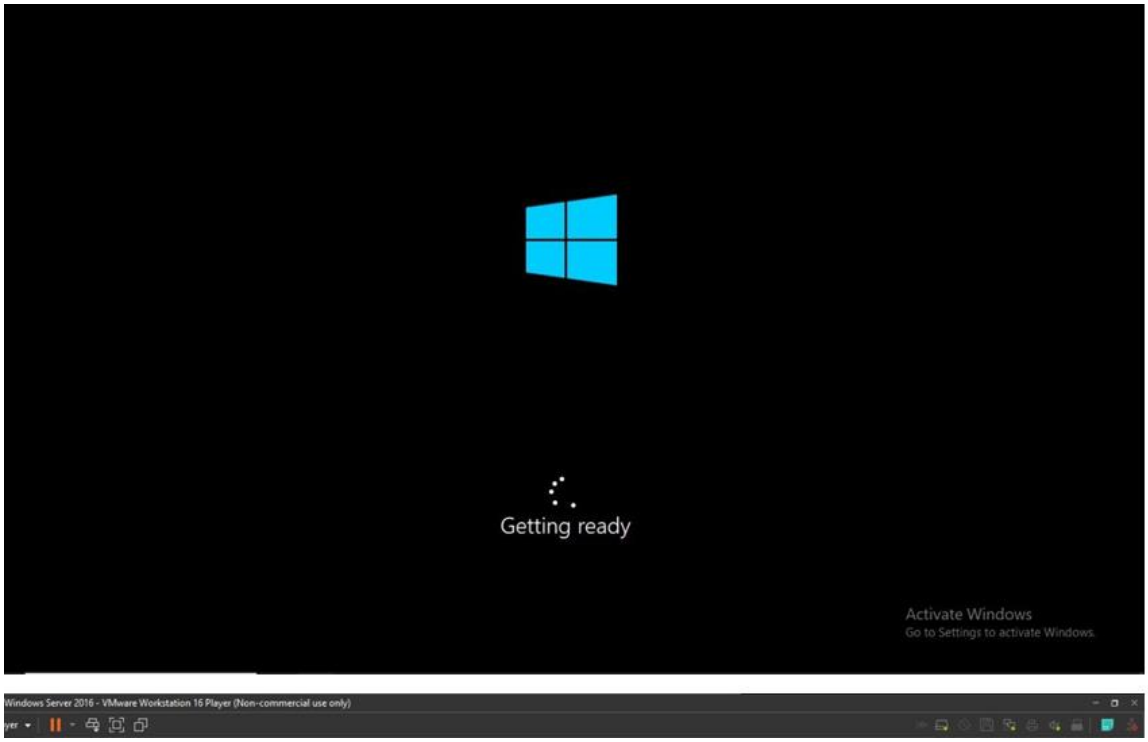
Increase attack surface,
When the control and data planes are separated, additional attack avenues are create d, such as rogue apps interacting with the controller through public APIs.

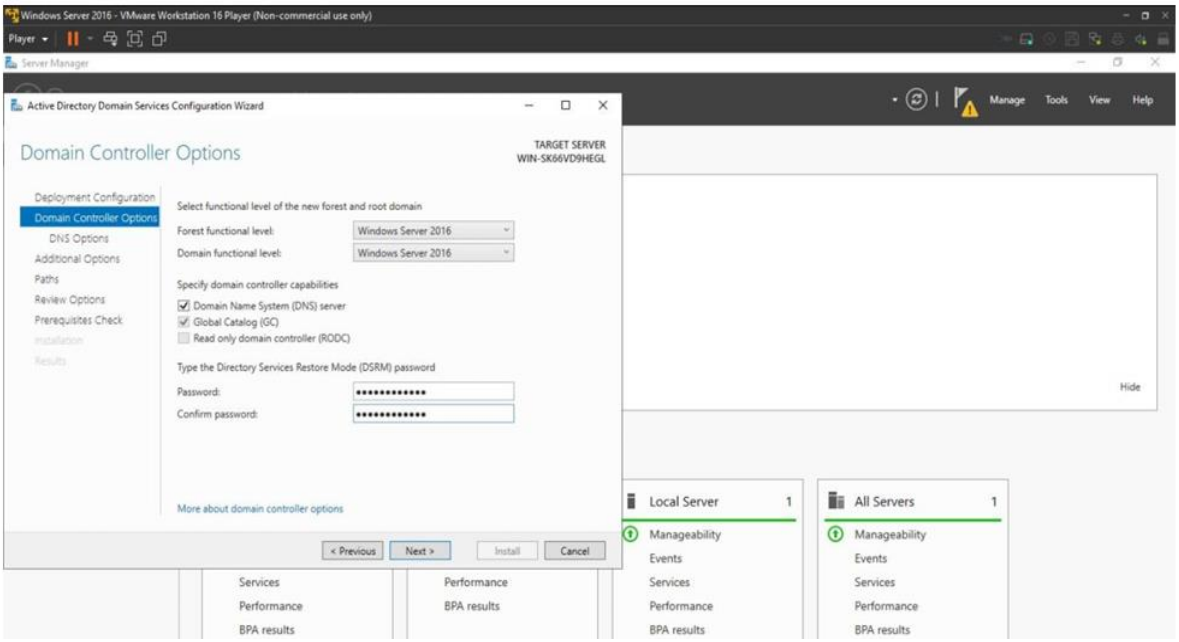
SDN relies on centralized control plane which if attacked might become single point of failure.

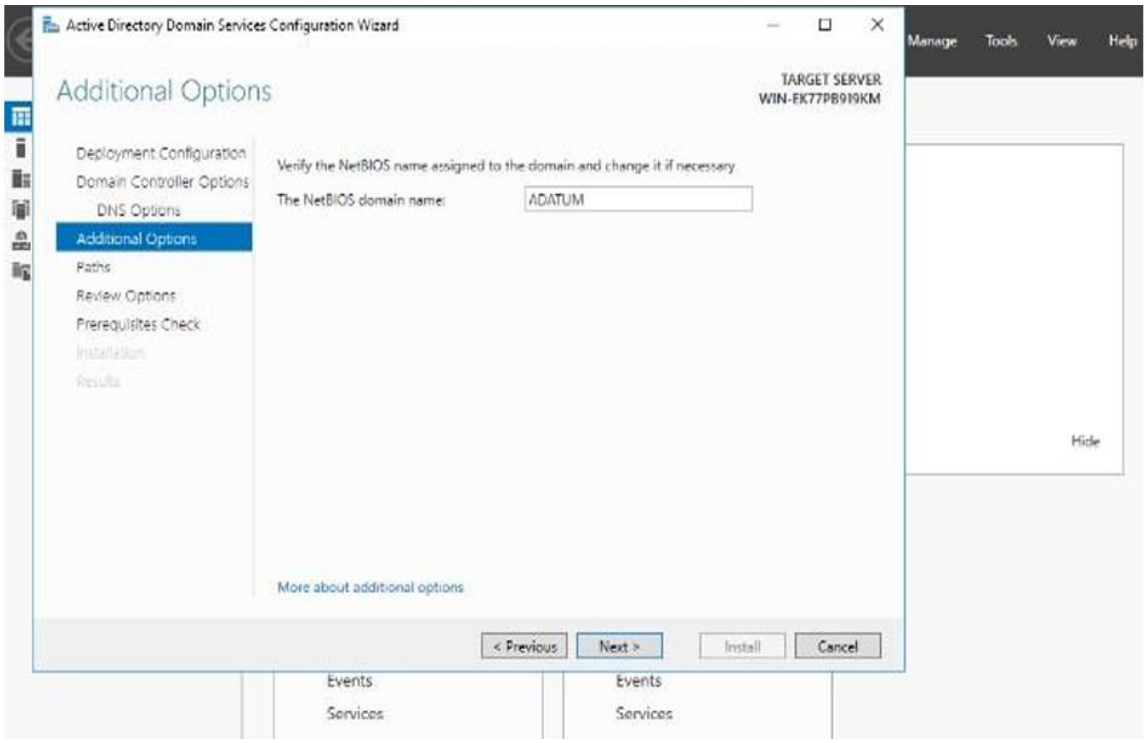
QUESTION 2

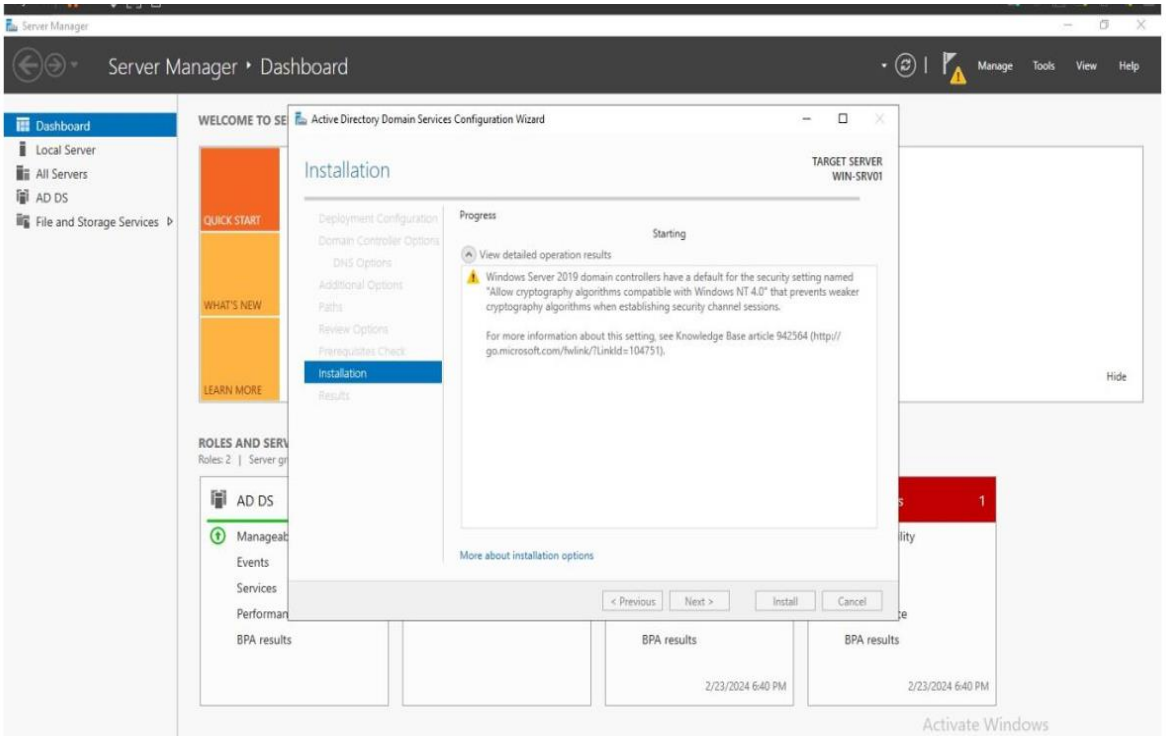

GNS3 2.2.29
GNS3 server version: 2.2.29
Release channel: 2.2
VM version: 0.12.1
Ubuntu version: focal
Qemu version: 4.2.1
Virtualization: vmware
KVM support available: True
Uptime: up 0 minutes
IP: 192.168.186.134 PORT: 80
To log in using SSH: ssh gns3@192.168.186.134
Password: gns3
To launch the Web-Ui: http://192.168.186.134
Images and projects are stored in '/opt/gns3'
< OK >
Activate Windows


File Edit View Control Node Annotate Tools Help
Switches
ATM switch
Ethernet hub
Ethernet switch
Frame Relay switch
WindowsServer2019CASP+-1
e0
WindowsServer2016CASP+-1
e0
R1
f0/0
f0/1
Switch1
e0
e1
e2
Windows10CASP+-1
e0
Topology Summary
Node
Console
R1
telnet 192.168.64.134...
Switch1
none
Windows10CASP+-1
none
WindowsServer2016CASP+-1
none
WindowsServer2019CASP+-1
none
Console

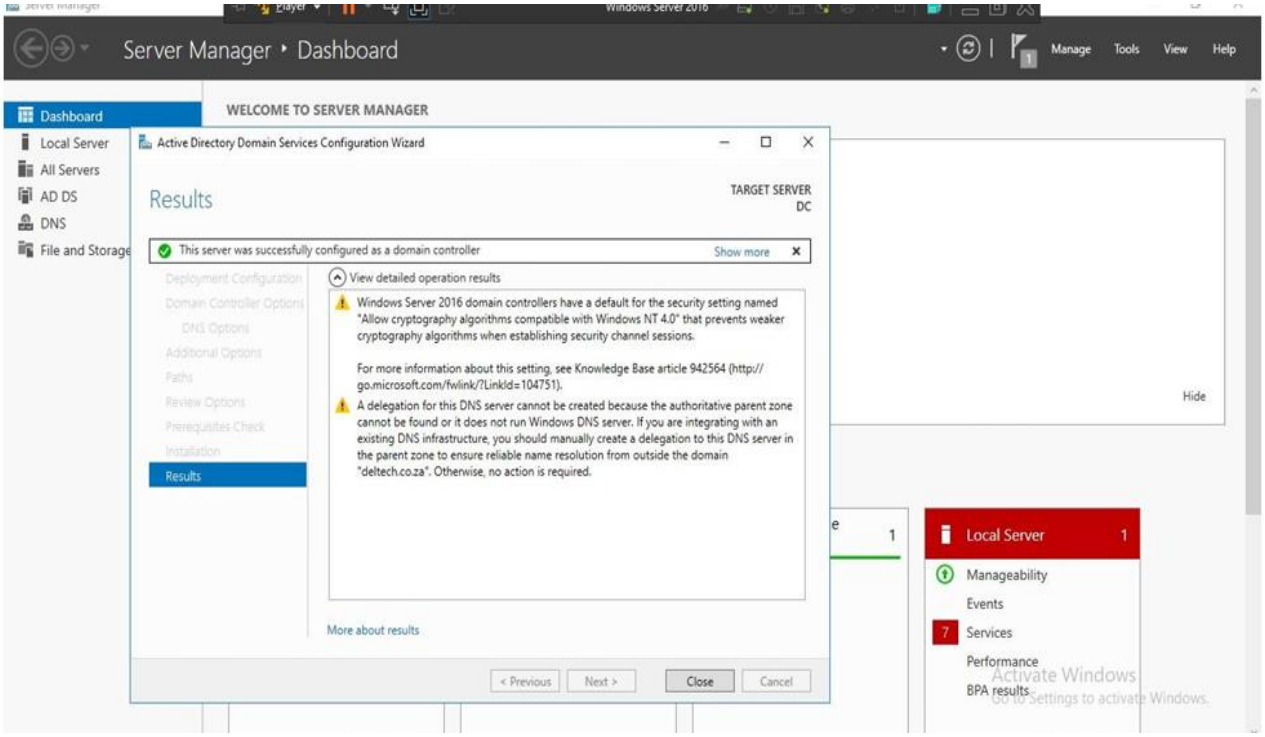

Getting ready
Activate Windows
Go to Settings to activate Windows.
Windows Server 2016 - VMware Workstation 16 Player (Non-commercial use only)

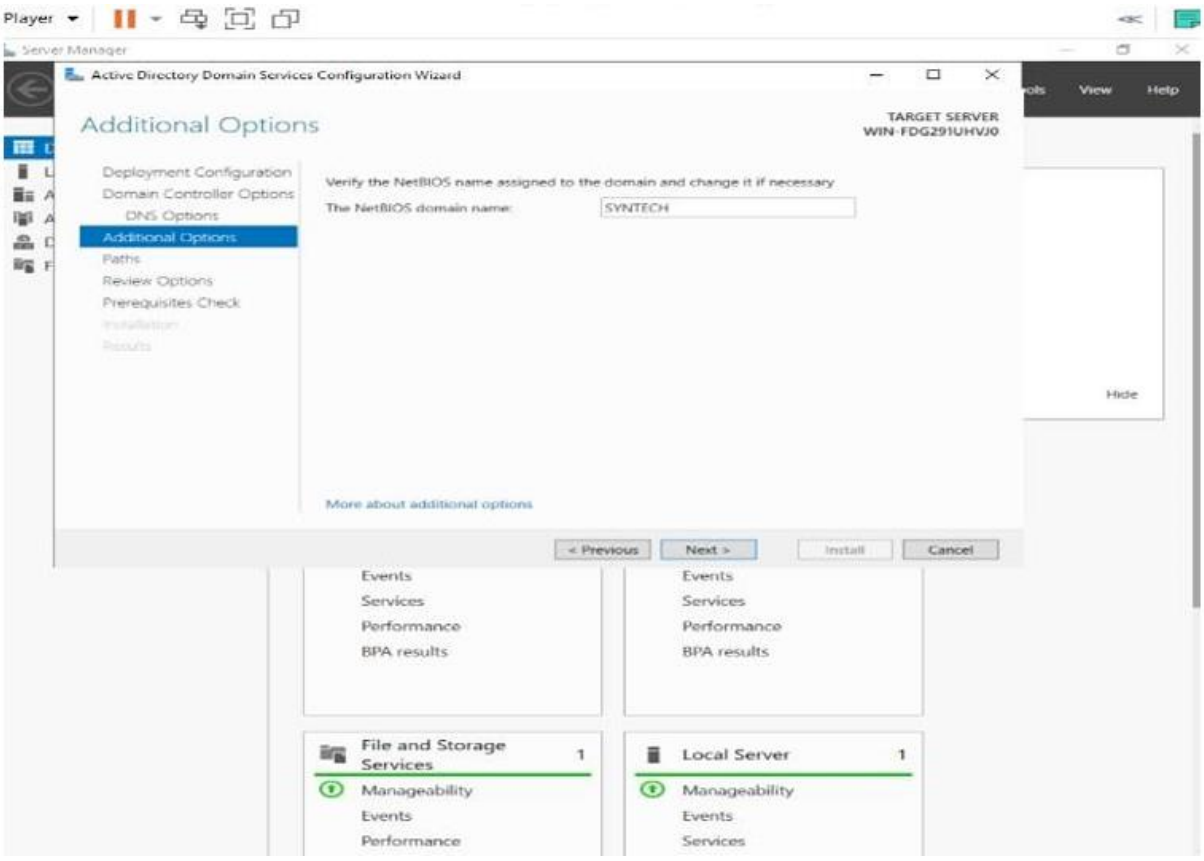

Windows Server 2016 - VMware Workstation 16 Player (Non-commercial use only)
Player
Server Manager
Manage
Tools
View
Help
Active Directory Domain Services Configuration Wizard
Domain Controller Options
TARGET SERVER
WIN-SK66VD9HEGL
Deployment Configuration
Domain Controller Options
DNS Options
Additional Options
Paths
Review Options
Prerequisites Check
Installation
Results
Select functional level of the new forest and root domain
Forest functional level:
Windows Server 2016
Domain functional level:
Windows Server 2016
Specify domain controller capabilities
Domain Name System (DNS) server
Global Catalog (GC)
Read only domain controller (RODC)
Type the Directory Services Restore Mode (DSRM) password
Password:
Confirm password:
More about domain controller options
< Previous
Next >
Install
Cancel
Hide
Local Server
1
Manageability
Events
Services
Performance
BPA results
All Servers
1
Manageability
Events
Services
Performance
BPA results
Services
Performance
BPA results
Performance
BPA results

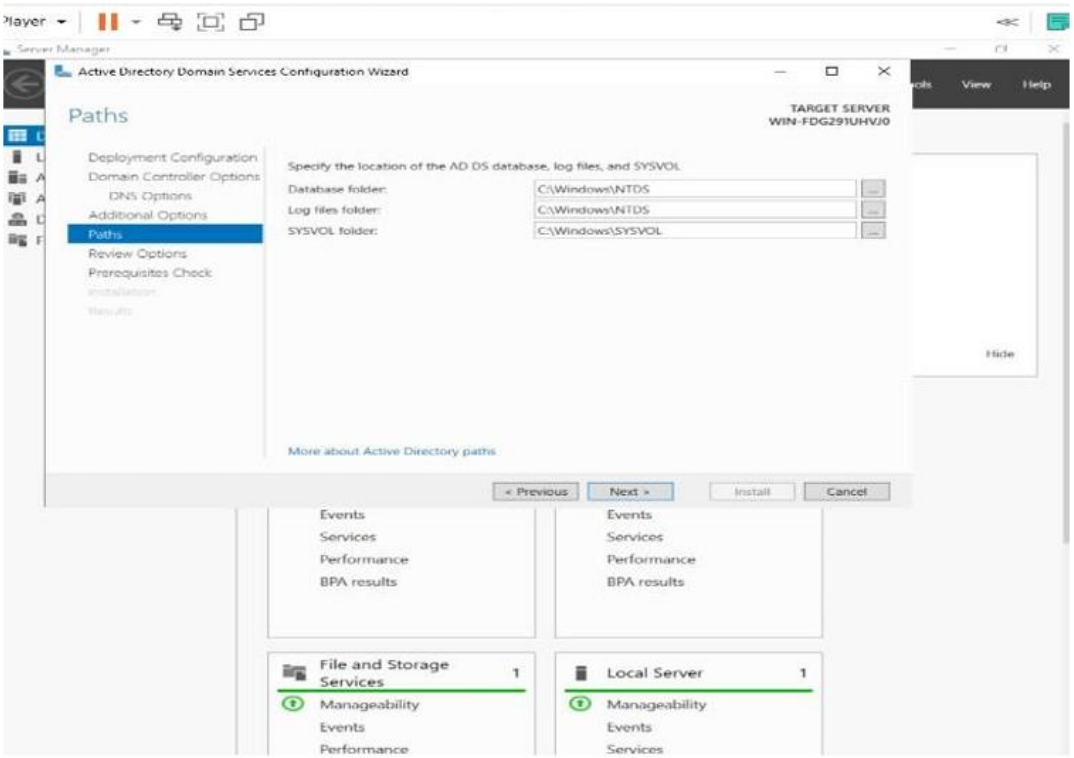

Active Directory Domain Services Configuration Wizard
Manage
Tools
View
Help
Additional Options
TARGET SERVER
WIN-FK77PB919KM
Deployment Configuration
Domain Controller Options
DNS Options
Additional Options
Paths
Review Options
Prerequisites Check
Installation
Results
Verify the NetBIOS name assigned to the domain and change it if necessary
The NetBIOS domain name:
ADATUM
More about additional options
< Previous
Next >
Install
Cancel
Hide
Events
Services
Events
Services

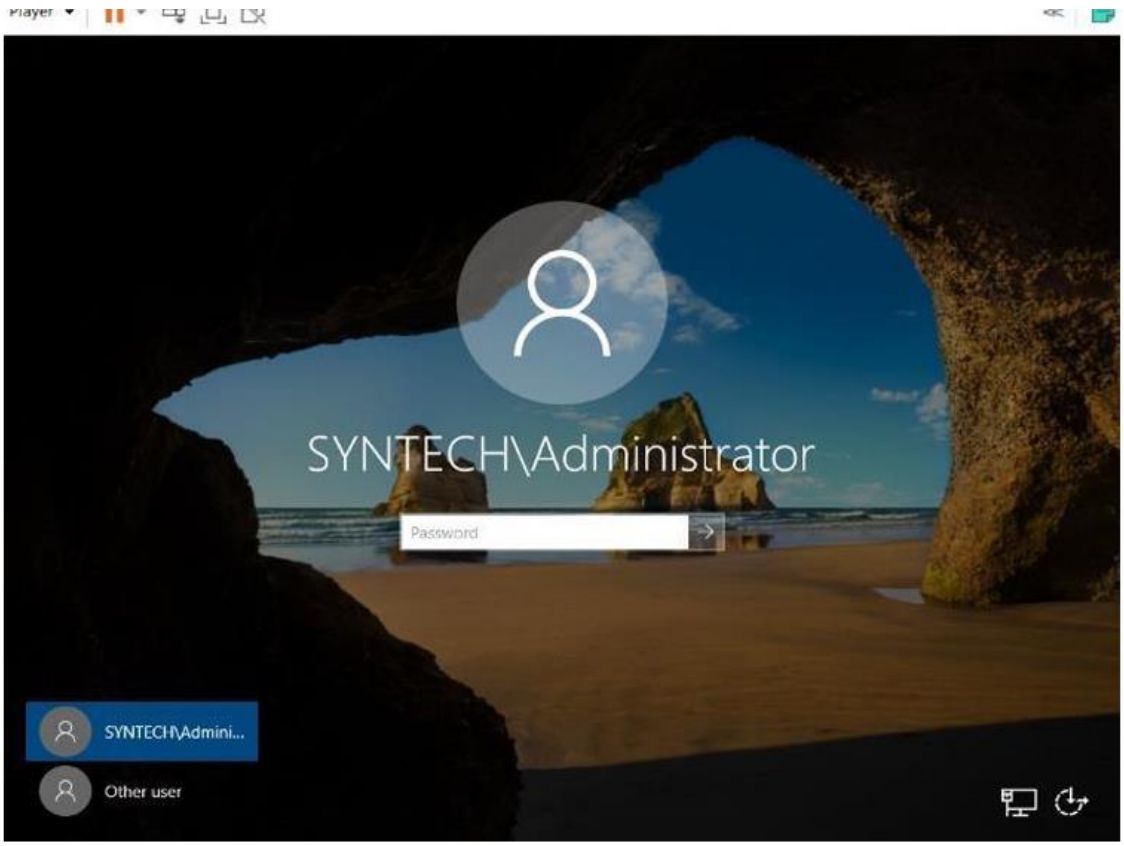

Server Manager
Server Manager • Dashboard
Manage
Tools
View
Help
Dashboard
Local Server
All Servers
AD DS
File and Storage Services
WELCOME TO SE
QUICK START
WHAT'S NEW
LEARN MORE
Hide
ROLES AND SERV
AD DS
Manageab
Events
Services
Performan
BPA results
BPA results
2/23/2024 6:40 PM
BPA results
2/23/2024 6:40 PM
Activate Windows
Active Directory Domain Services Configuration Wizard
Installation
TARGET SERVER
WIN-SRV01
Deployment Configuration
Domain Controller Options
DNS Options
Additional Options
Paths
Review Options
Prerequisites Check
Installation
Results
Progress
Starting
View detailed operation results
Windows Server 2019 domain controllers have a default for the security setting named "Allow cryptography algorithms compatible with Windows NT 4.0" that prevents weaker cryptography algorithms when establishing security channel sessions.
For more information about this setting, see Knowledge Base article 942564 (http://go.microsoft.com/fwlink/?LinkId=104751).
More about installation options
< Previous
Next >
Install
Cancel

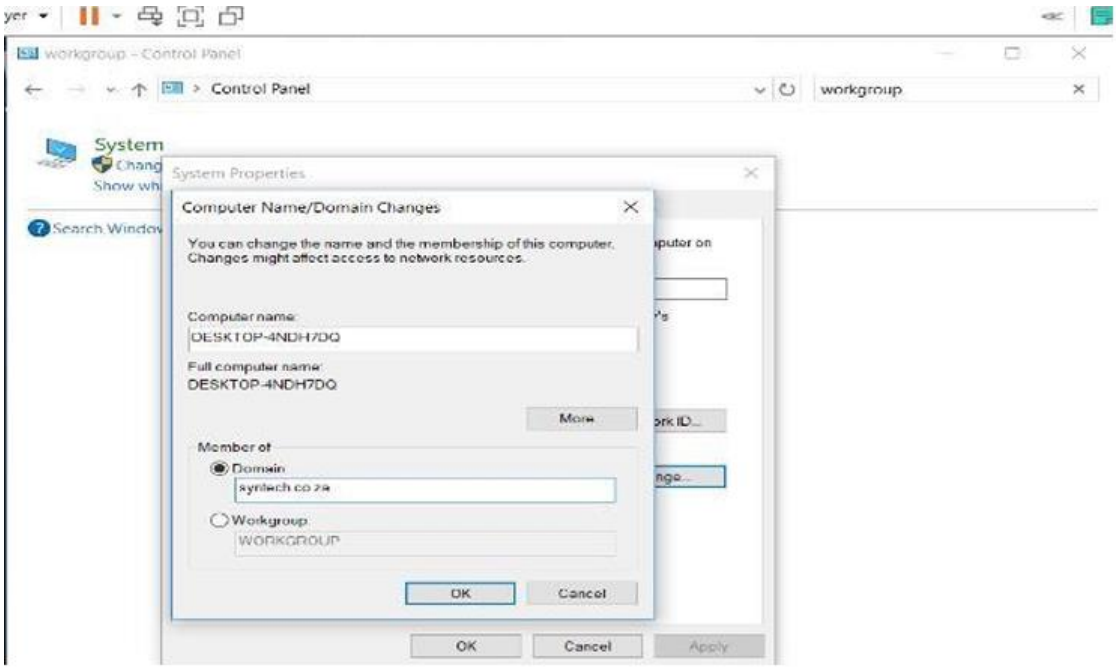
ADATUM\Administrator
Password
ADATUM\Admins...

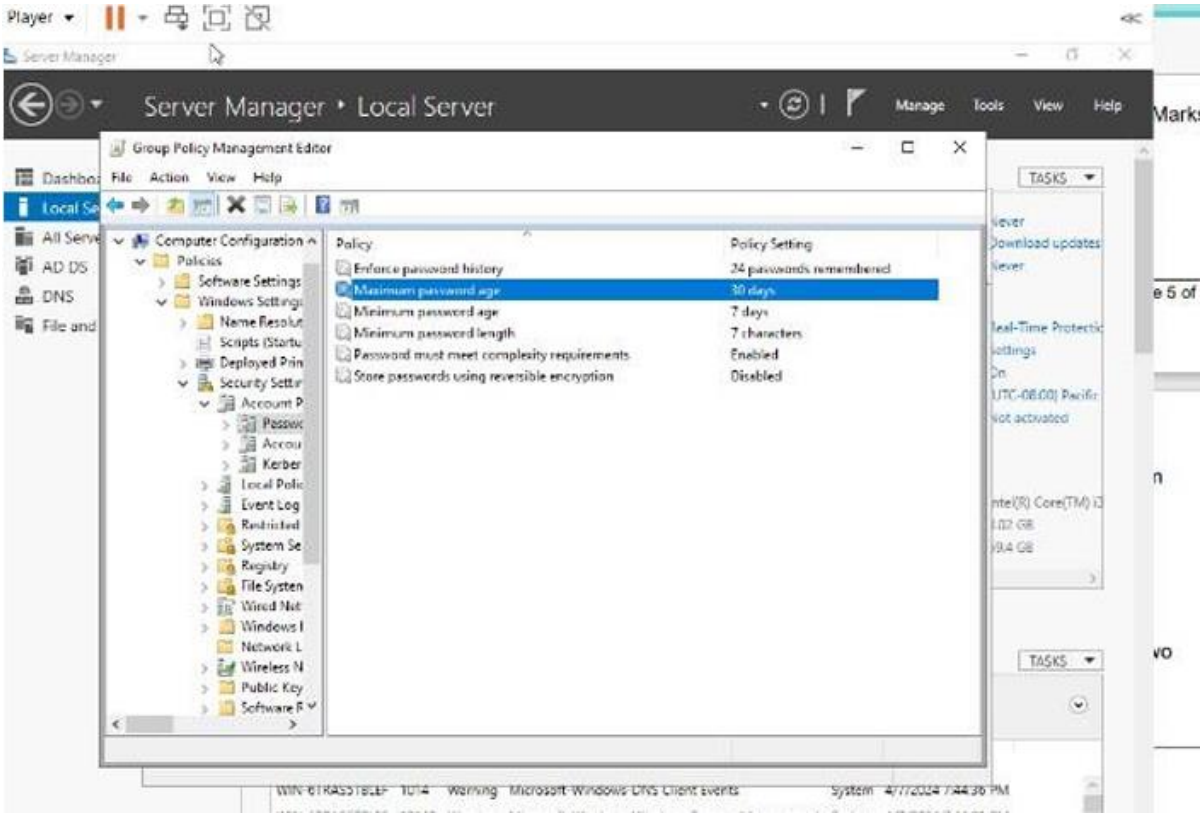

Server Manager ▸ Dashboard
Manage
Tools
View
Help
Dashboard
Local Server
All Servers
AD DS
DNS
File and Storage
WELCOME TO SERVER MANAGER
Active Directory Domain Services Configuration Wizard
Results
TARGET SERVER
DC
This server was successfully configured as a domain controller
Show more
View detailed operation results
Windows Server 2016 domain controllers have a default for the security setting named "Allow cryptography algorithms compatible with Windows NT 4.0" that prevents weaker cryptography algorithms when establishing security channel sessions.
For more information about this setting, see Knowledge Base article 942564 (http://go.microsoft.com/fwlink/?LinkId=104751).
A delegation for this DNS server cannot be created because the authoritative parent zone cannot be found or it does not run Windows DNS server. If you are integrating with an existing DNS infrastructure, you should manually create a delegation to this DNS server in the parent zone to ensure reliable name resolution from outside the domain "deltech.co.za". Otherwise, no action is required.
Deployment Configuration
Domain Controller Options
DNS Options
Additional Options
Paths
Review Options
Prerequisites Check
Installation
Results
More about results
< Previous
Next >
Close
Cancel
Hide
Local Server
Manageability
Events
Services
Performance
BPA results
Activate Windows
Go to Settings to activate Windows.

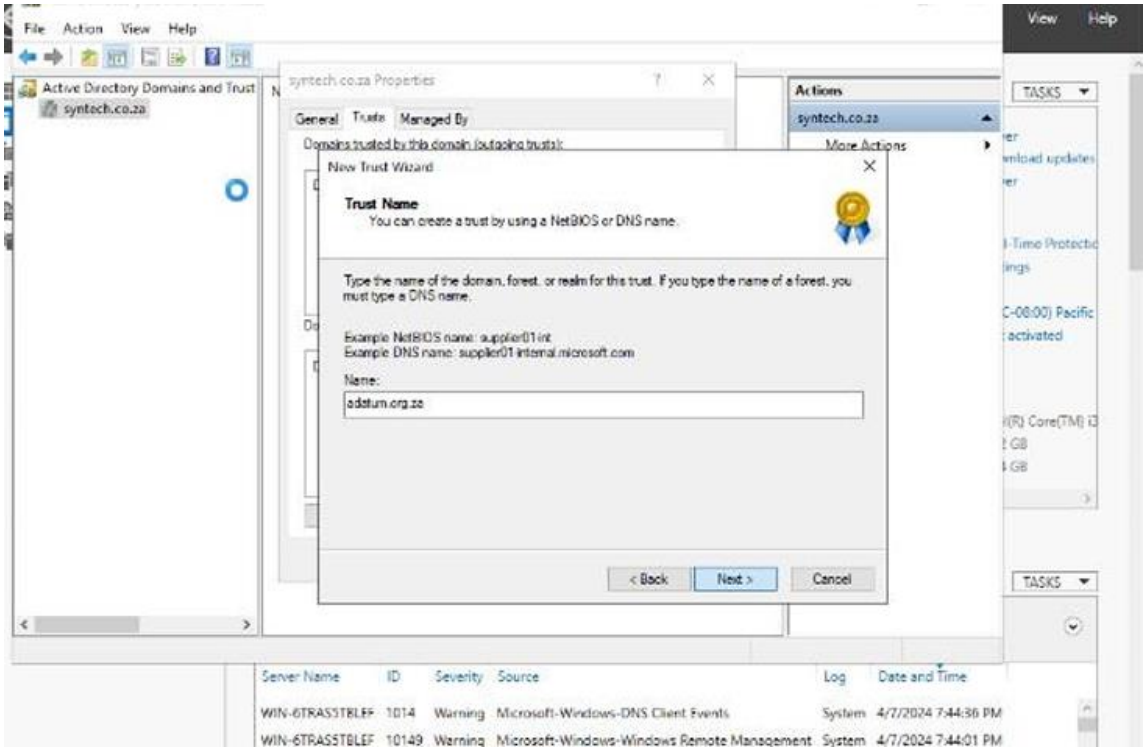

Player
Server Manager
Active Directory Domain Services Configuration Wizard
Additional Options
TARGET SERVER
WIN-FDG291UHVJ0
Deployment Configuration
Domain Controller Options
DNS Options
Additional Options
Paths
Review Options
Prerequisites Check
Verify the NetBIOS name assigned to the domain and change it if necessary
The NetBIOS domain name:
SYNTECH
More about additional options
< Previous
Next >
Install
Cancel
Events
Services
Performance
BPA results
File and Storage Services
Manageability
Events
Performance
Local Server
Manageability
Events
Services
Hide

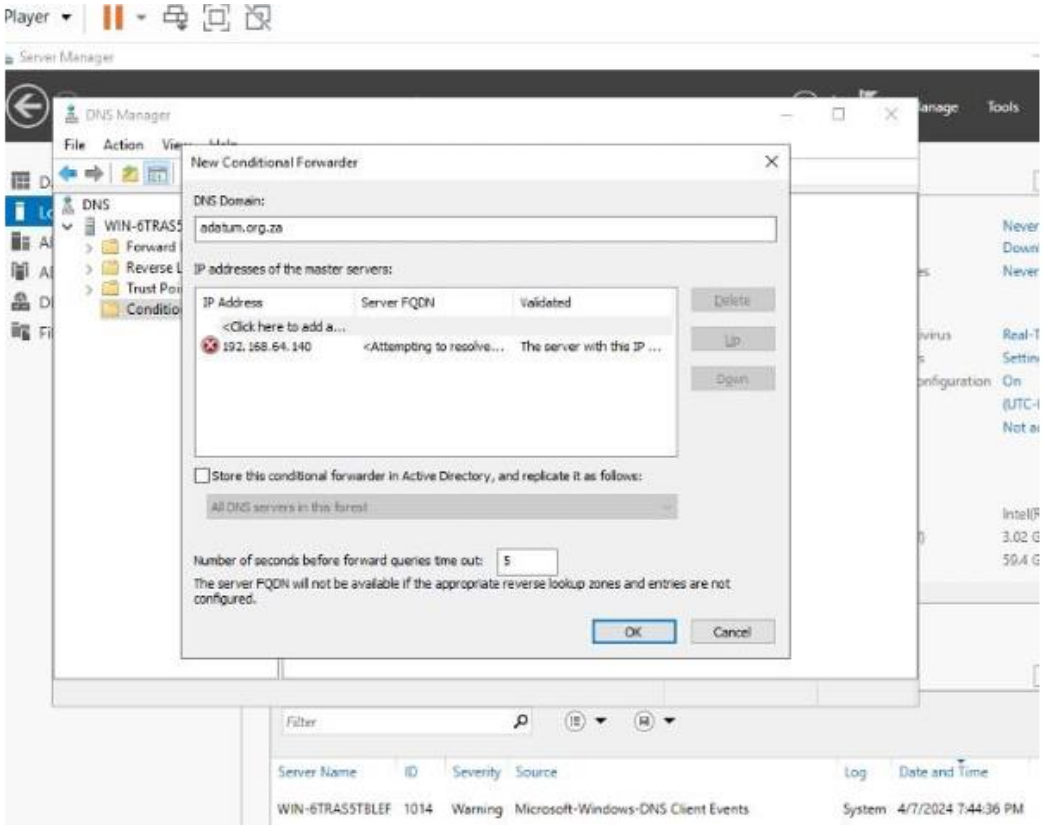

Active Directory Domain Services Configuration Wizard
Paths
TARGET SERVER
WIN-FDG291UHVJ0
Deployment Configuration
Domain Controller Options
DNS Options
Additional Options
Paths
Review Options
Prerequisites Check
Specify the location of the AD DS database, log files, and SYSVOL
Database folder:
C:\Windows\NTDS
Log files folder:
C:\Windows\NTDS
SYSVOL folder:
C:\Windows\SYSVOL
More about Active Directory paths
< Previous
Next >
Install
Cancel
Events
Services
Performance
BPA results
File and Storage Services
Manageability
Local Server

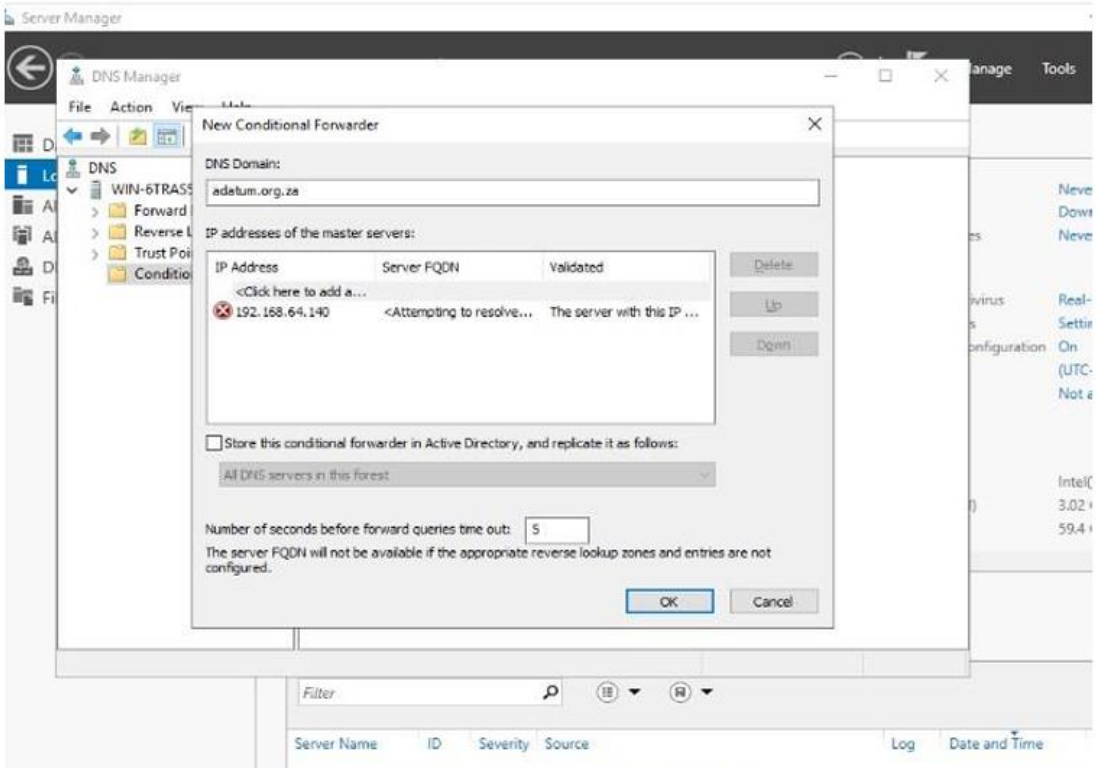

SYNTECH\Administrator
Password
SYNTECH\Admini...
Other user

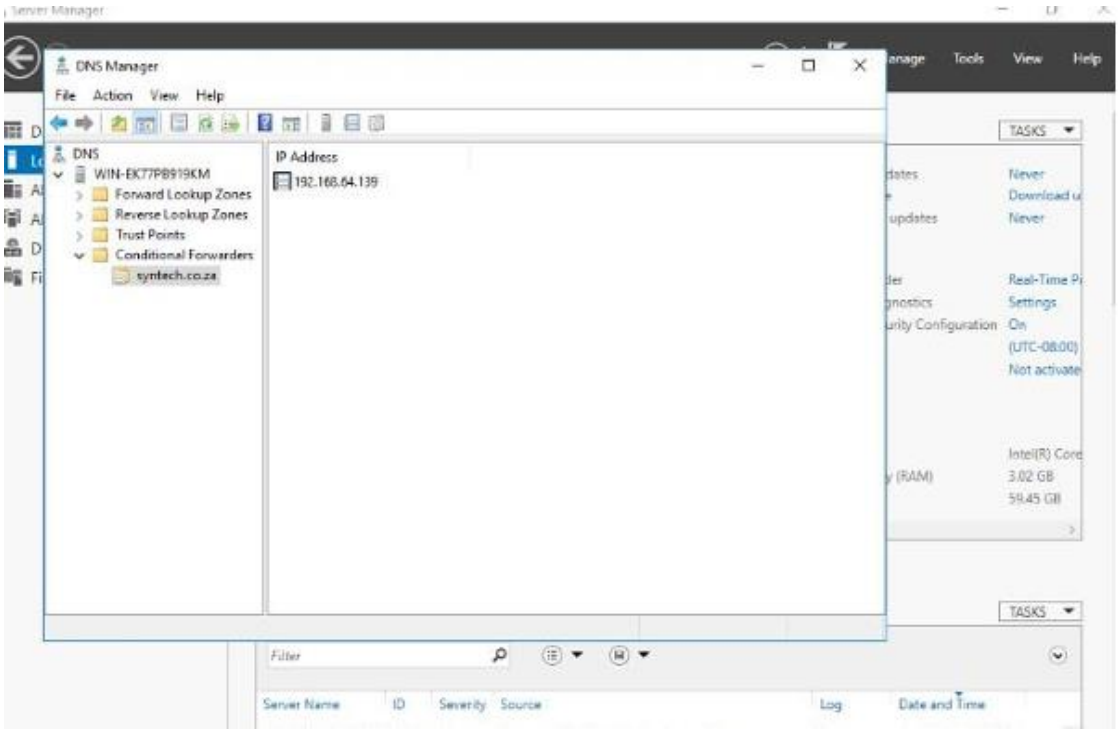

workgroup - Control Panel
Control Panel
workgroup
System
System Properties
Computer Name/Domain Changes
You can change the name and the membership of this computer. Changes might affect access to network resources.
Computer name:
DESKTOP-4NDH7DQ
Full computer name:
DESKTOP-4NDH7DQ
More...
Member of
Domain:
Workgroup:
WORKGROUP
OK
Cancel
OK
Cancel
Apply

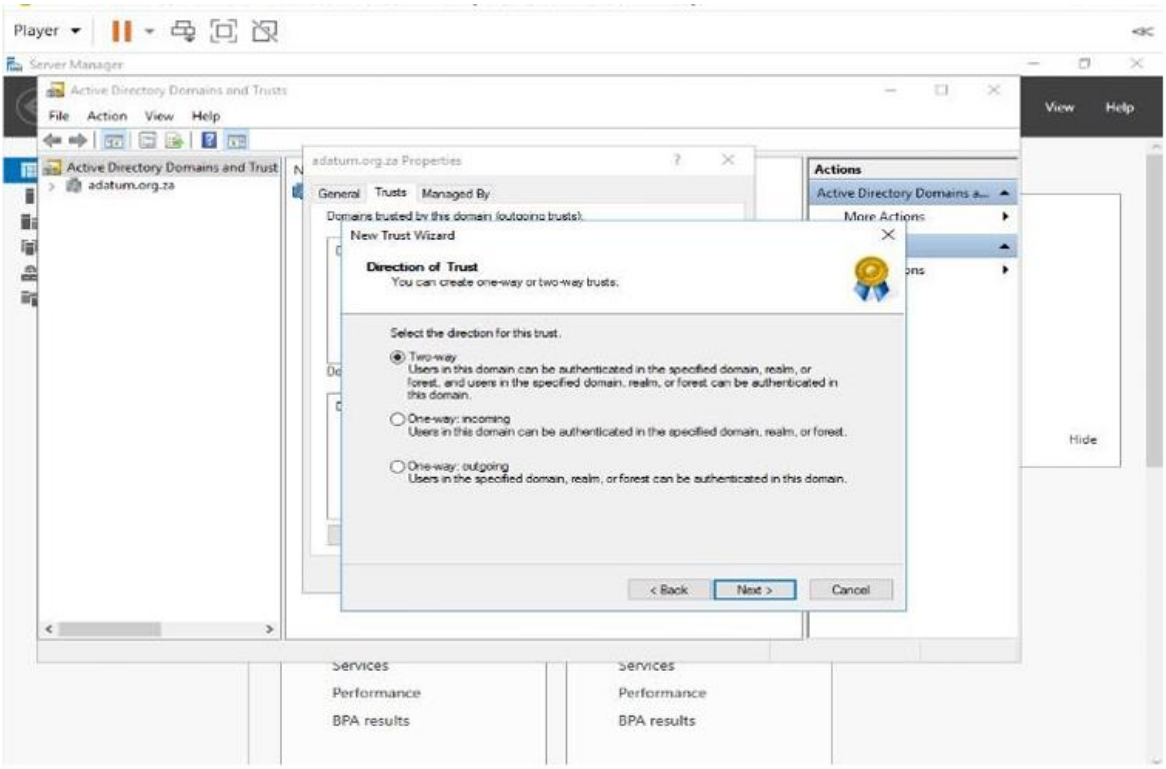

Server Manager
Server Manager • Local Server
Manage
Tools
View
Help
Group Policy Management Editor
File
Action
View
Help
Computer Configuration
Policies
Software Settings
Windows Settings
Policy
Policy Setting
Enforce password history
24 passwords remembered
Maximum password age
30 days
Minimum password age
7 days
Minimum password length
7 characters
Password must meet complexity requirements
Enabled
Store passwords using reversible encryption
Disabled
TASKS

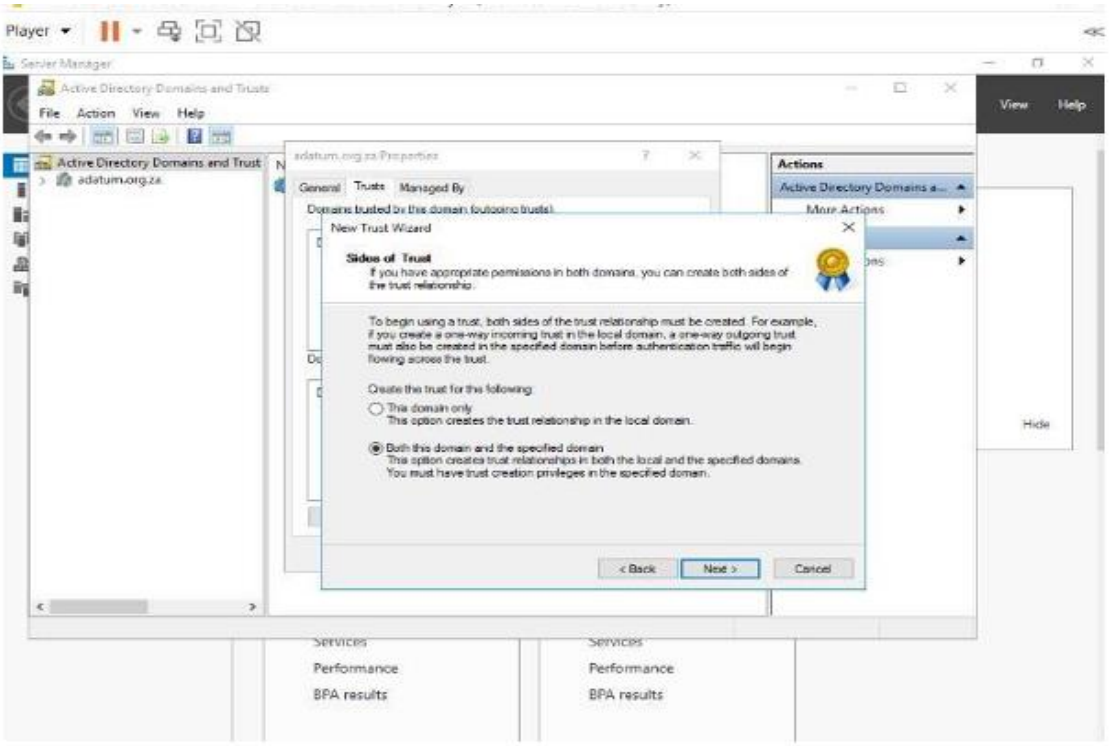

File Action View Help
Active Directory Domains and Trust
syntech.co.za
syntech.co.za Properties
General Trusts Managed By
Domains trusted by this domain (outgoing trusts):
New Trust Wizard
Trust Name
You can create a trust by using a NetBIOS or DNS name.
Type the name of the domain, forest, or realm for this trust. If you type the name of a forest, you must type a DNS name.
Example NetBIOS name: supplier01 int
Example DNS name: supplier01.internal.microsoft.com
Name:
adatum.org.za
< Back
Next >
Cancel
Actions
syntech.co.za
More Actions
TASKS
Server Name ID Severity Source Log Date and Time
WIN-6TRAS5TBLEF 1014 Warning Microsoft-Windows-DNS Client Events System 4/7/2024 7:44:36 PM
WIN-6TRAS5TBLEF 10149 Warning Microsoft-Windows-Windows Remote Management System 4/7/2024 7:44:01 PM

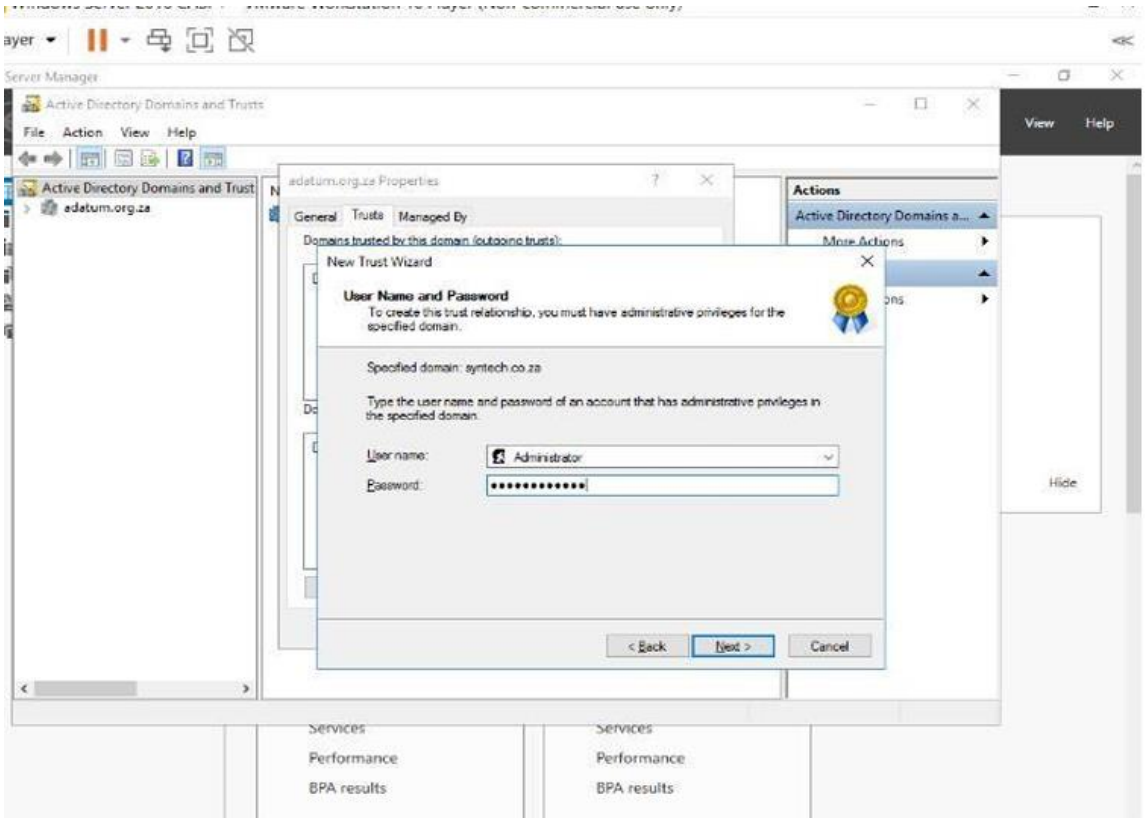

Player
Server Manager
DNS Manager
File Action View Help
DNS
WIN-6TRAS5
Forward
Reverse L
Trust Poi
Conditio
New Conditional Forwarder
DNS Domain:
adatum.org.za
IP addresses of the master servers:
IP Address
Server FQDN
Validated
<Click here to add a...
192.168.64.140
<Attempting to resolve...
The server with this IP ...
Delete
Up
Down
Store this conditional forwarder in Active Directory, and replicate it as follows:
All DNS servers in this forest
Number of seconds before forward queries time out:
5
The server FQDN will not be available if the appropriate reverse lookup zones and entries are not configured.
OK
Cancel
Filter
Server Name ID Severity Source Log Date and Time
WIN-6TRAS5TBLEF 1014 Warning Microsoft-Windows-DNS Client Events System 4/7/2024 7:44:36 PM

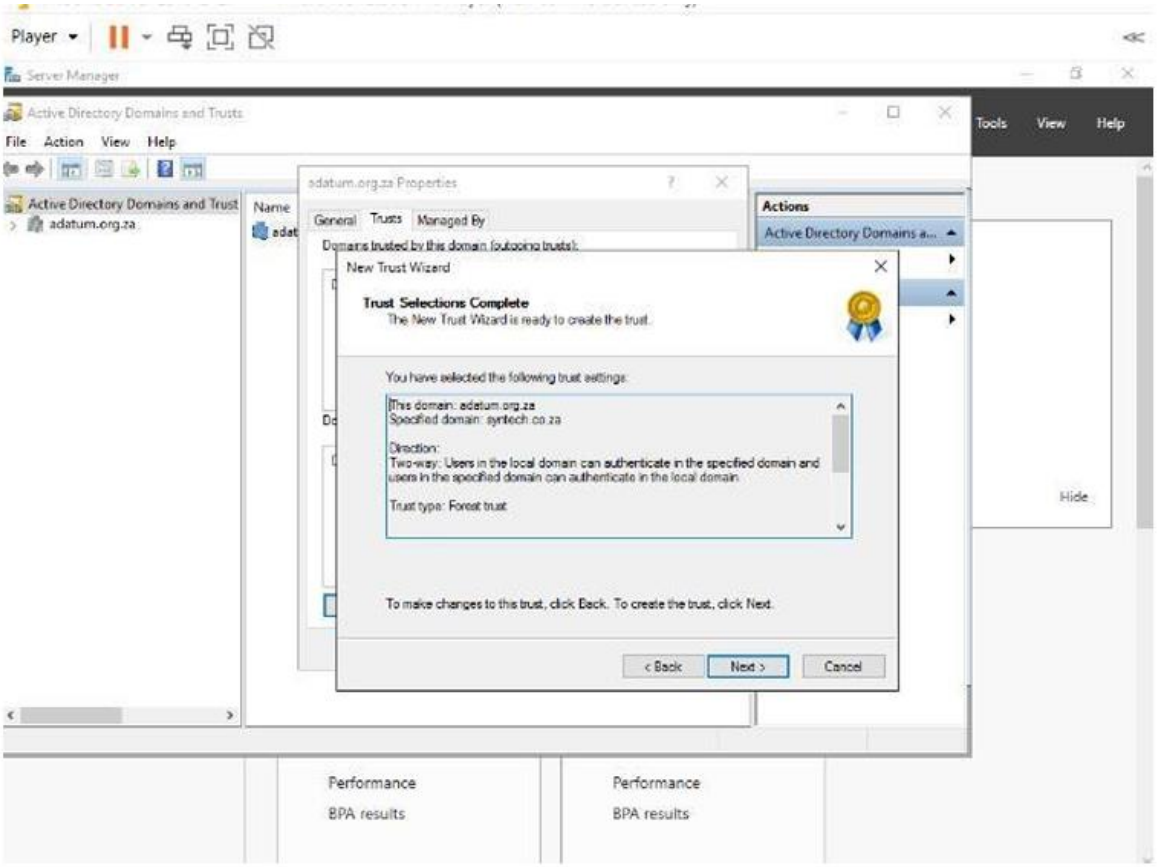

Server Manager
DNS Manager
File Action View Help
New Conditional Forwarder
DNS Domain:
adatum.org.za
IP addresses of the master servers:
IP Address
Server FQDN
Validated
<Click here to add a...
192.168.64.140
<Attempting to resolve...
The server with this IP ...
Delete
Up
Down
Store this conditional forwarder in Active Directory, and replicate it as follows:
All DNS servers in this forest
Number of seconds before forward queries time out:
5
The server FQDN will not be available if the appropriate reverse lookup zones and entries are not configured.
OK
Cancel
DNS
WIN-6TRAS5
Forward
Reverse
Trust Poi
Conditio
Filter
Server Name
ID
Severity
Source
Log
Date and Time

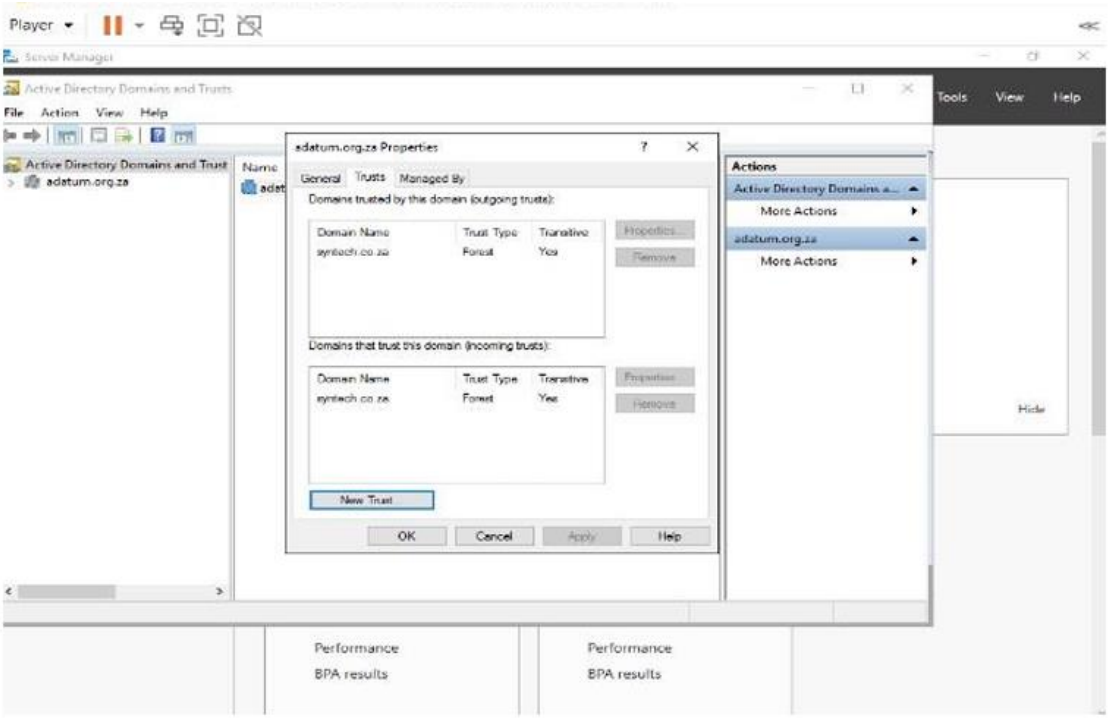

Server Manager
DNS Manager
File Action View Help
DNS
WIN-EK77PB919KM
Forward Lookup Zones
Reverse Lookup Zones
Trust Points
Conditional Forwarders
syntech.co.za
IP Address
192.168.64.139
TASKS
Filter
Server Name
ID
Severity
Source
Log
Date and Time


Player
Server Manager
Active Directory Domains and Trusts
File Action View Help
View Help
Active Directory Domains and Trust
adatum.org.za
adatum.org.za Properties
General Trusts Managed By
Domains trusted by this domain (outgoing trusts).
Actions
Active Directory Domains a...
More Actions
Hide
New Trust Wizard
Direction of Trust
You can create one-way or two-way trusts.
Select the direction for this trust.
Two-way
Users in this domain can be authenticated in the specified domain, realm, or forest, and users in the specified domain, realm, or forest can be authenticated in this domain.
One-way: incoming
Users in this domain can be authenticated in the specified domain, realm, or forest.
One-way: outgoing
Users in the specified domain, realm, or forest can be authenticated in this domain.
< Back
Next >
Cancel
Services
Performance
BPA results


Player
Server Manager
Active Directory Domains and Trusts
File Action View Help
View Help
Active Directory Domains and Trust
adatum.org.za
adatum.org.za Properties
General Trusts Managed By
Domains trusted by this domain (outgoing trusts).
Actions
Active Directory Domains a...
More Actions
Hide
New Trust Wizard
Sides of Trust
If you have appropriate permissions in both domains, you can create both sides of the trust relationship.
To begin using a trust, both sides of the trust relationship must be created. For example, if you create a one-way incoming trust in the local domain, a one-way outgoing trust must also be created in the specified domain before authentication traffic will begin flowing across the trust.
Create the trust for the following:
This domain only
This option creates the trust relationship in the local domain.
Both this domain and the specified domain
This option creates trust relationships in both the local and the specified domains. You must have trust creation privileges in the specified domain.
< Back
Next >
Cancel
Services
Performance
BPA results


Server Manager
Active Directory Domains and Trusts
File Action View Help
View Help
Active Directory Domains and Trust
adatum.org.za
adatum.org.za Properties
General Trusts Managed By
Domains trusted by this domain (outgoing trusts):
New Trust Wizard
User Name and Password
To create this trust relationship, you must have administrative privileges for the specified domain.
Specified domain: syntech.co.za
Type the user name and password of an account that has administrative privileges in the specified domain.
User name: Administrator
Password:
< Back Next > Cancel
Actions
Active Directory Domains a...
More Actions
Hide
Services
Performance
BPA results


Player
Server Manager
Active Directory Domains and Trusts
File Action View Help
Tools View Help
Active Directory Domains and Trust
adatum.org.za
Name
adat
adatum.org.za Properties
General Trusts Managed By
Domains trusted by this domain (outgoing trusts):
New Trust Wizard
Trust Selections Complete
The New Trust Wizard is ready to create the trust.
You have selected the following trust settings:
This domain: adatum.org.za
Specified domain: syntech.co.za
Direction:
Two-way: Users in the local domain can authenticate in the specified domain and users in the specified domain can authenticate in the local domain.
Trust type: Forest trust
To make changes to this trust, click Back. To create the trust, click Next.
< Back Next > Cancel
Actions
Active Directory Domains a...
Hide
Performance
BPA results
Windows Server 2016 CASP+ - VMware Workstation 16 Player (Non-commercial use only)


Player
Server Manager
Active Directory Domains and Trusts
File Action View Help
Tools View Help
Active Directory Domains and Trust
adatum.org.za
Name
adat
adatum.org.za Properties
General Trusts Managed By
Domains trusted by this domain (outgoing trusts):
Domain Name
Trust Type
Transitive
Forest
Yes
Properties...
Remove
Domains that trust this domain (incoming trusts):
Domain Name
Trust Type
Transitive
Forest
Yes
Properties
Remove
New Trust
OK
Cancel
Apply
Help
Actions
Active Directory Domains a...
More Actions
adatum.org.za
More Actions
Hide
Performance
BPA results
Performance
BPA results